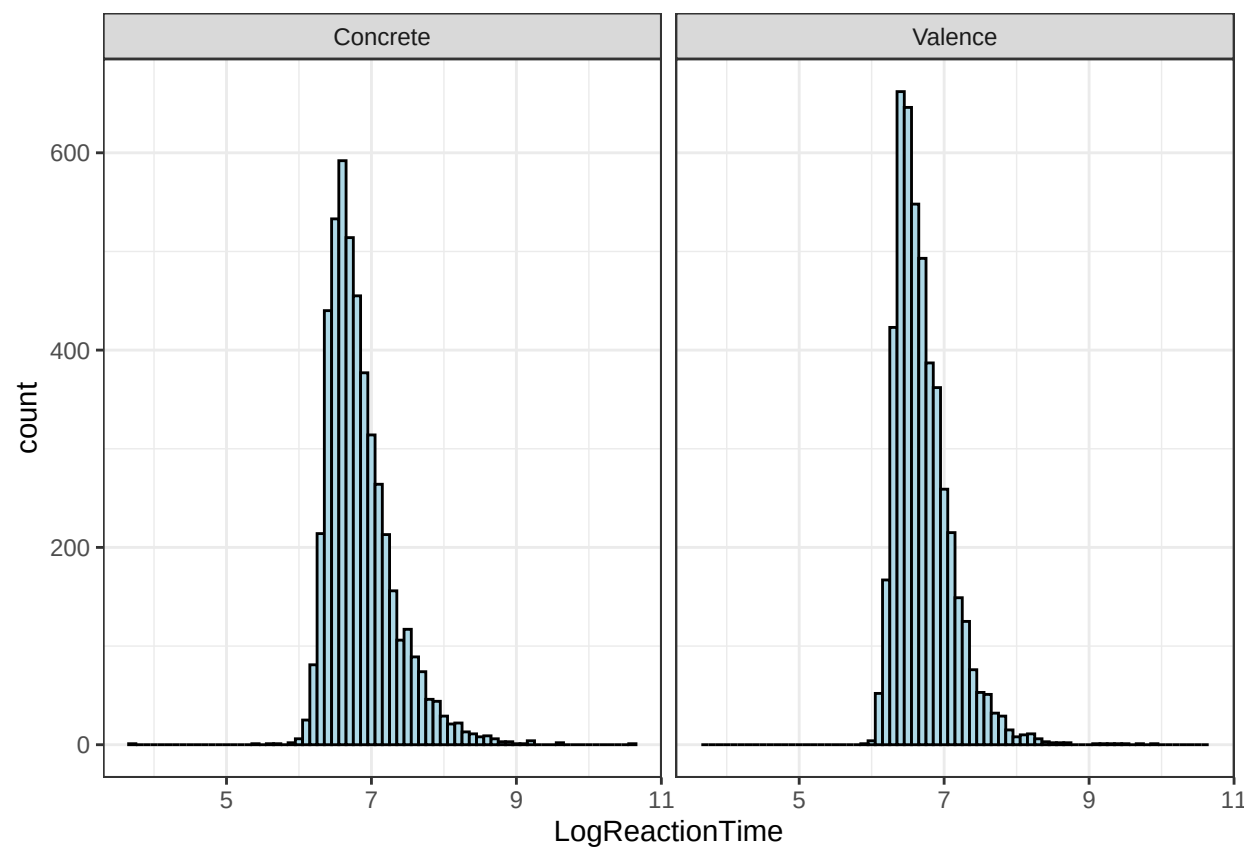


Verbs Conc-Abs: Analysis

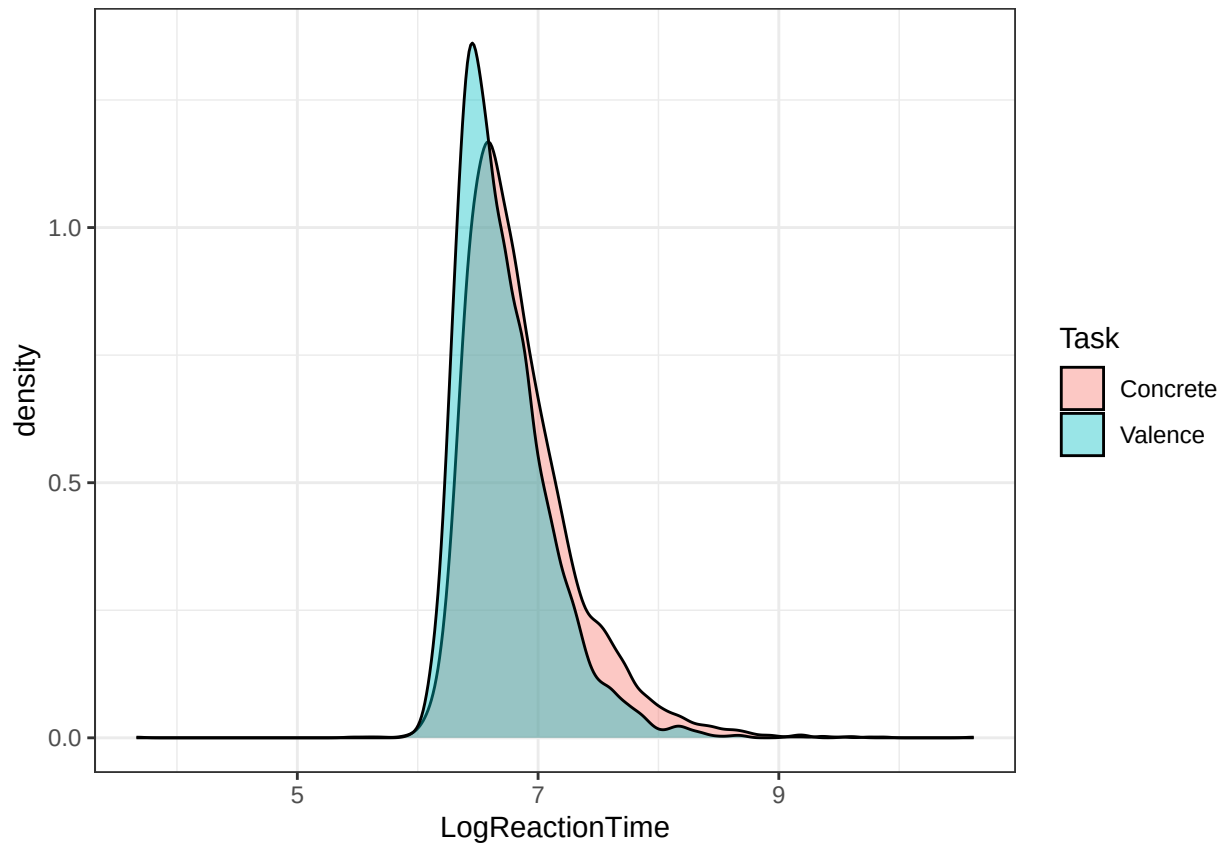
author

2026-05-29

```
ggplot(d, aes(x=LogReactionTime)) +  
  geom_histogram(binwidth = .1, fill = "lightblue", color = "black") +  
  facet_wrap(~Task)
```



```
ggplot(d, aes(x=LogReactionTime, fill=Task)) +  
  geom_density(alpha = .4)
```



```
names(d)
```

```
## [1] "X"           "ID.true"      "Word"         "Label"
## [5] "ConcValCombo" "Task"         "BlockOrder"   "Group"
## [9] "Response"     "Accuracy"     "EventTime"    "Value"
## [13] "RT"           "ReactionTime" "Key_value_F"  "Key_value_J"
## [17] "Comments"     "LogReactionTime" "LogRT"        "TrialNumber"
```

```
dcen <- d %>%
```

```
  mutate(Word = as.factor(Word),
         ID.true = as.factor(ID.true),
         Task = as.factor(Task),
         cAccuracy = as.numeric(Accuracy) - mean(as.numeric(Accuracy)),
         cTask = as.numeric(Task) - mean(as.numeric(Task)))
```

```
m <- lmer(LogReactionTime ~ cAccuracy*cTask + (1+cTask|Word) + (1+cTask|ID.true), data = dcen)
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cAccuracy * cTask + (1 + cTask | Word) + (1 +
##       cTask | ID.true)
## Data: dcen
##
## REML criterion at convergence: 7409.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
```

```

## -9.8359 -0.5999 -0.1593  0.4131  9.5533
##
## Random effects:
##   Groups   Name                Variance Std.Dev. Corr
##   Word      (Intercept) 0.003690 0.06075
##             cTask      0.006214 0.07883  0.00
##   ID.true   (Intercept) 0.047166 0.21718
##             cTask      0.028544 0.16895 -0.08
##   Residual                0.120935 0.34776
## Number of obs: 9600, groups:  Word, 40; ID.true, 40
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      6.78049    0.03583   44.93308 189.216 < 2e-16 ***
## cAccuracy        -0.14525    0.01678  9520.87440  -8.654 < 2e-16 ***
## cTask            -0.13001    0.03033   52.61728  -4.287 7.76e-05 ***
## cAccuracy:cTask    0.12953    0.03355  9520.62607   3.860 0.000114 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cAccrc cTask
## cAccuracy    -0.003
## cTask        -0.063 -0.022
## cAccrcy:cTs  -0.009  0.310 -0.007

```

```

agr <- d %>%
  group_by(Word,Task) %>%
  summarize(MeanAccuracy = mean(Accuracy),
            MeanReactionTime = mean(ReactionTime))

```

```

## `summarise()` has grouped output by 'Word'. You can override using the
## `.groups` argument.

```

```

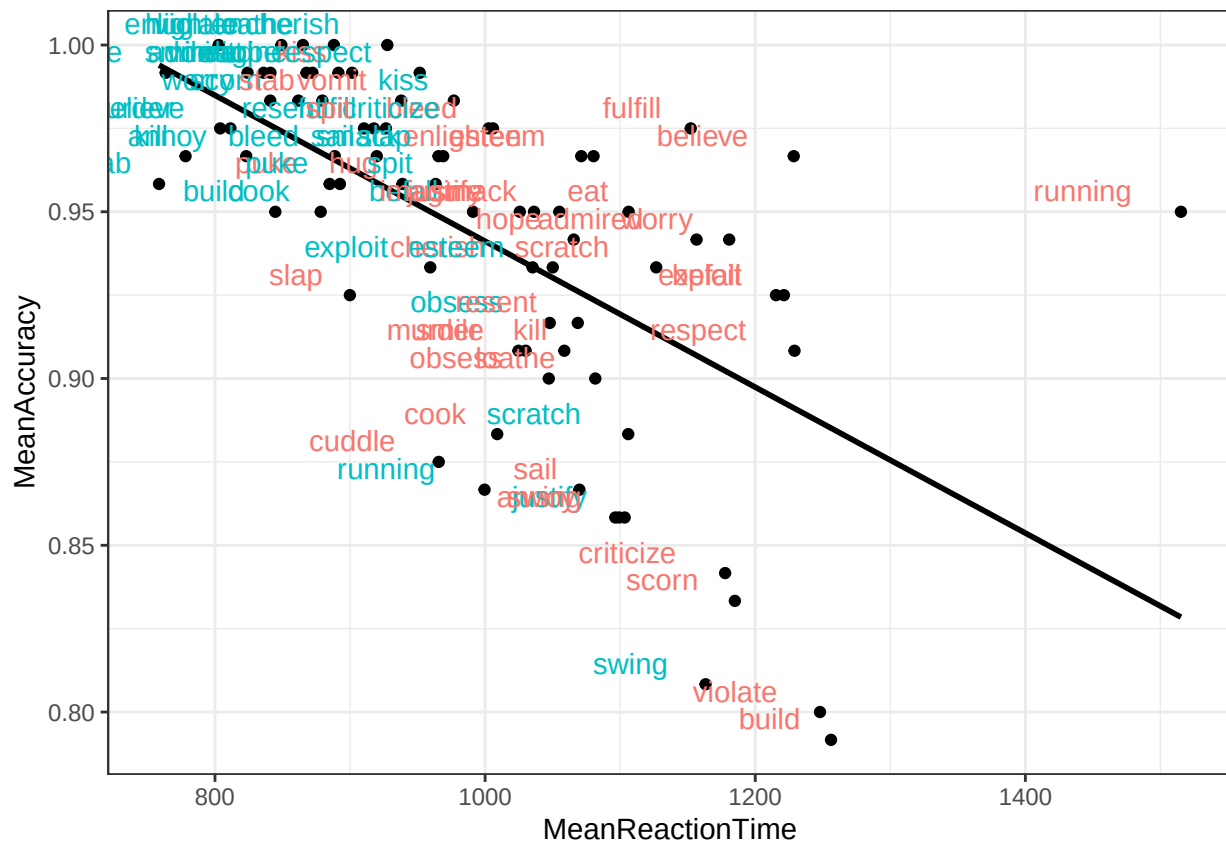
ggplot(agr, aes(x = MeanReactionTime, y = MeanAccuracy)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  geom_text(aes(label = Word, color = Task), vjust = -0.5, hjust = 1.5) +
  # guides(legend = "none")
  theme(legend.position = "none") # Remove the legend

```

```

## `geom_smooth()` using formula = 'y ~ x'

```



```
# ggsave("../graphs/exp1b_accXrt.pdf",width = 5, height = 3)
```

```
# Compute highest accuracy for Concrete
concrete_accuracy <- d %>%
  group_by(Word,Task) %>%
  summarize(MeanAccuracy = mean(Accuracy),
            MeanReactionTime = mean(ReactionTime)) %>%
  filter(Task == "Concrete") %>%
  select(Word, MeanAccuracy) %>%
  rename(ConcreteAccuracy = MeanAccuracy) %>%
  arrange(desc(ConcreteAccuracy)) %>%
  head(10)
```

```
## `summarise()` has grouped output by 'Word'. You can override using the
## `.groups` argument.
```

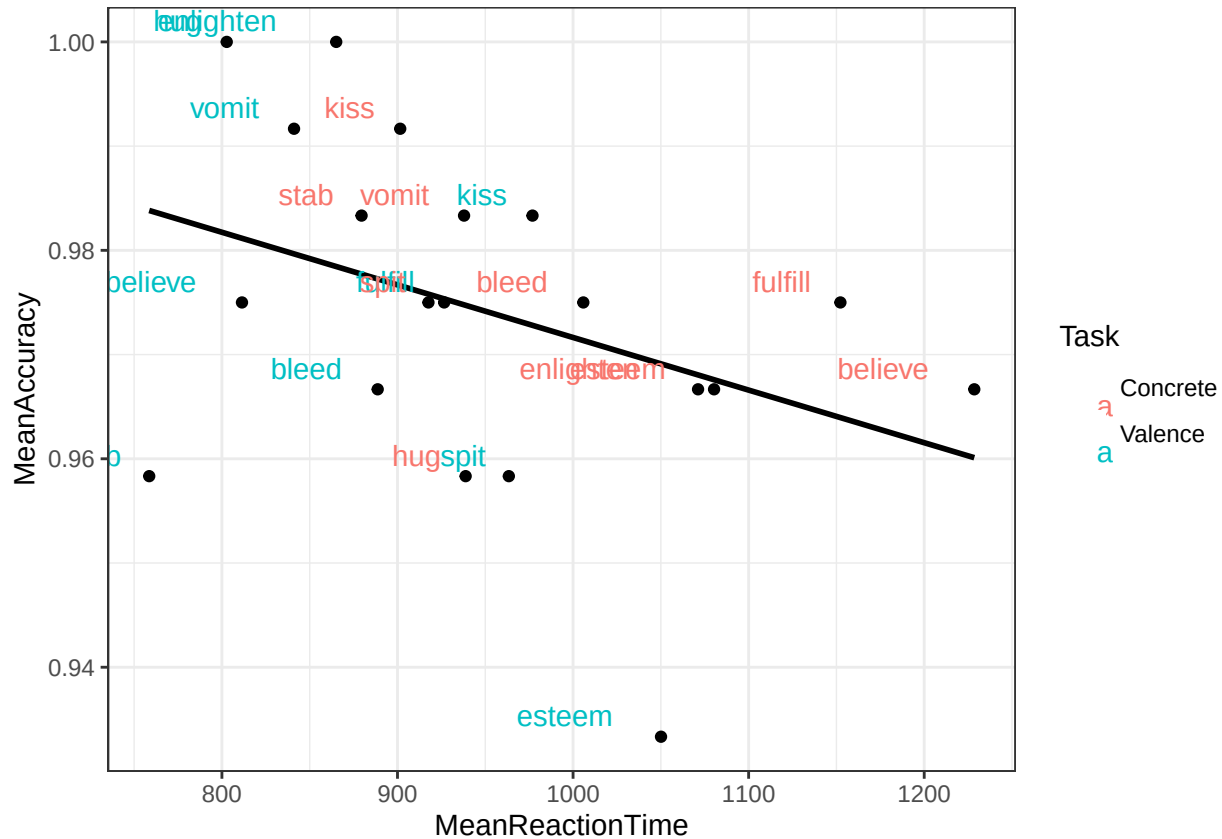
```
# View(concrete_accuracy)
```

```
agr <- d %>%
  filter(Word %in% concrete_accuracy$Word) %>%
  group_by(Word,Task) %>%
  summarize(MeanAccuracy = mean(Accuracy),
            MeanReactionTime = mean(ReactionTime))
```

```
## `summarise()` has grouped output by 'Word'. You can override using the
## `.groups` argument.
```

```
ggplot(agr, aes(x = MeanReactionTime, y = MeanAccuracy)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  geom_text(aes(label = Word, color = Task), vjust = -0.5, hjust = 1.5)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

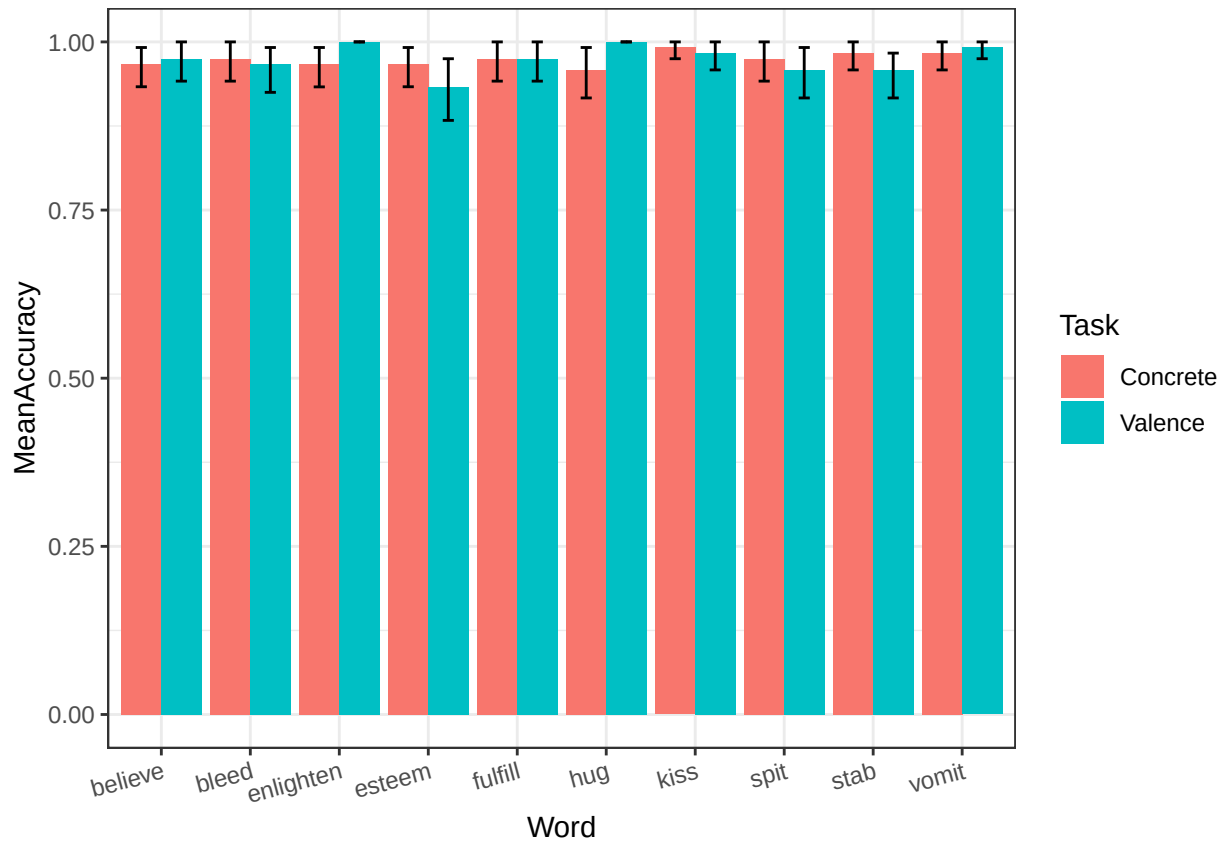


```
# guides(legend = "none")
# theme(legend.position = "none") # Remove the legend
ggsave("../graphs/exp1b_accXrt.pdf", width = 5, height = 3)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
agr <- d %>%
  filter(Word %in% concrete_accuracy$Word) %>%
  group_by(Word, Task) %>%
  reframe(MeanAccuracy = mean(Accuracy),
          CILow = ci.low(Accuracy),
          CIHigh = ci.high(Accuracy)) %>%
  mutate(YMin = MeanAccuracy - CILow,
          YMax = MeanAccuracy + CIHigh)
# View(agr)

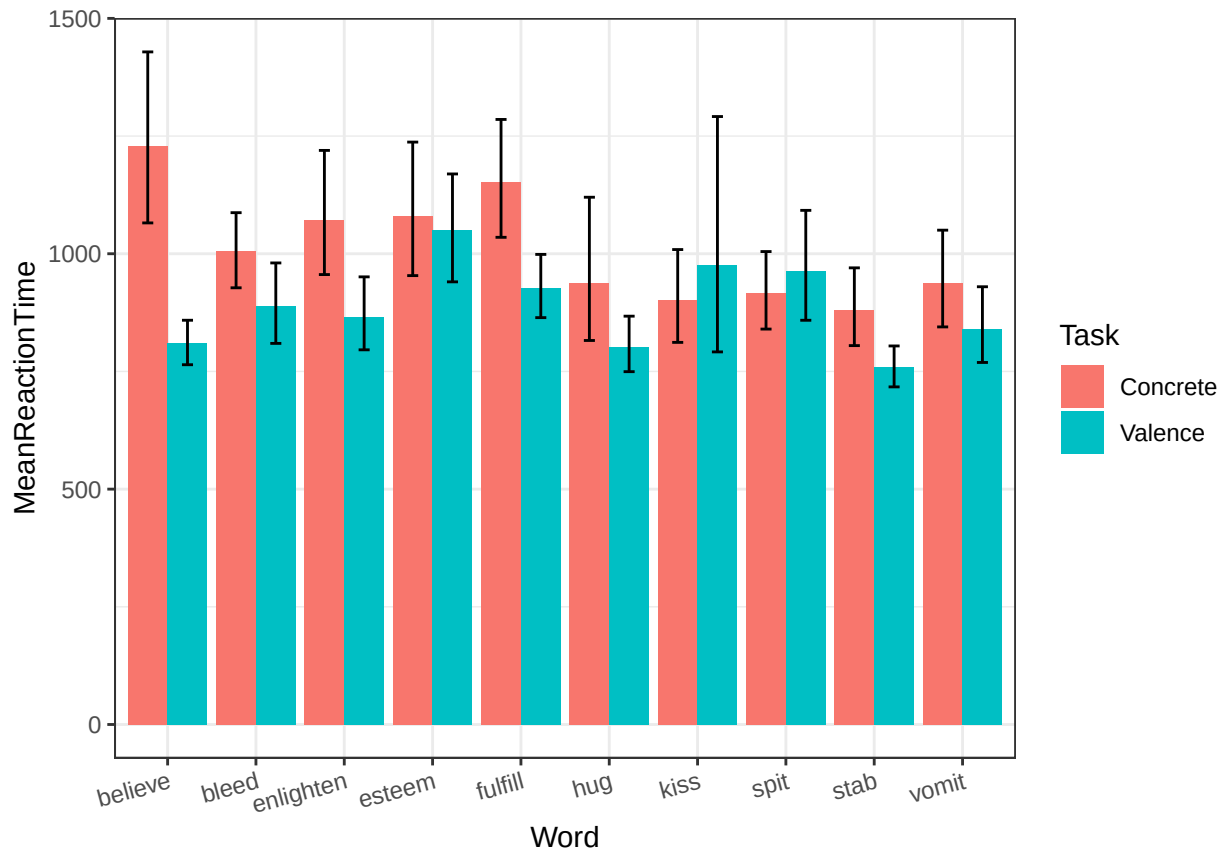
dodge = position_dodge(.9)
ggplot(data=agr, aes(x=Word, y=MeanAccuracy, fill = Task)) +
  geom_bar(position=dodge, stat="identity") +
  geom_errorbar(aes(ymin=YMin, ymax=YMax), width=.25, position=position_dodge(0.9)) +
  theme(axis.text.x = element_text(angle = 15, hjust = 1))
```



```
# ggsave("../graphs/exp1_mostacc_verbs_acc.pdf",width = 6, height = 3)
```

```
agr <- d %>%
  filter(Word %in% concrete_accuracy$Word) %>%
  group_by(Word,Task) %>%
  reframe(MeanReactionTime = mean(ReactionTime),
           CILow = ci.low(ReactionTime),
           CIHigh = ci.high(ReactionTime)) %>%
  mutate(YMin = MeanReactionTime - CILow,
         YMax = MeanReactionTime + CIHigh)
# View(agr)

dodge = position_dodge(.9)
ggplot(data=agr, aes(x=Word,y=MeanReactionTime, fill = Task)) +
  geom_bar(position=dodge,stat="identity") +
  geom_errorbar(aes(ymin=YMin,ymax=YMax),width=.25,position=position_dodge(0.9))+
  theme(axis.text.x = element_text(angle = 15, hjust = 1))
```



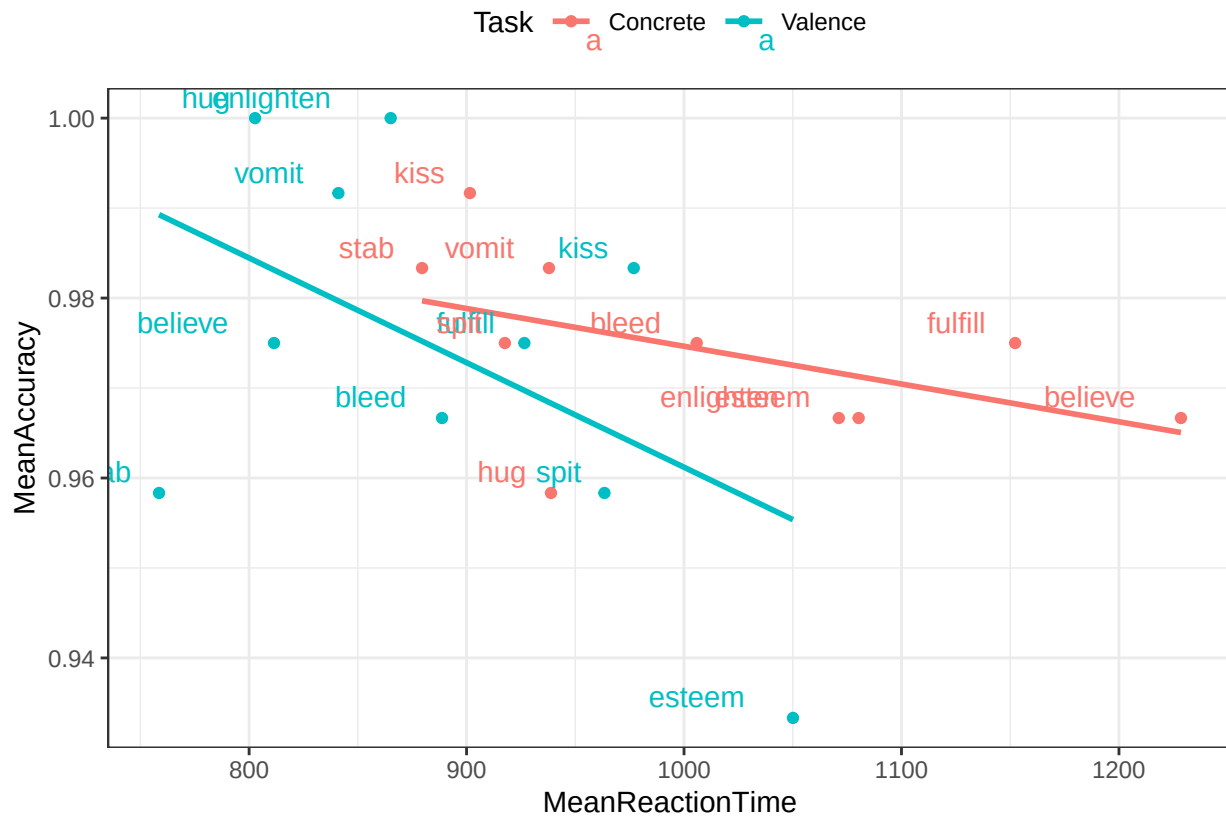
```
# ggsave("../graphs/exp1_mostacc_verbs_rt.pdf", width = 6, height = 3)
```

```
agr <- d %>%
  # mutate(Task = ifelse(Task == "Valence", "Valence", "Conceptual"),
  #           Task = factor(Task, levels = c("Valence", "Conceptual"))) %>%
  filter(Word %in% concrete_accuracy$Word) %>%
  filter(Word != "obsess") %>%
  group_by(Word, Task) %>%
  summarize(MeanAccuracy = mean(Accuracy),
            MeanReactionTime = mean(ReactionTime))
```

```
## `summarise()` has grouped output by 'Word'. You can override using the
## `.groups` argument.
```

```
ggplot(agr, aes(x = MeanReactionTime, y = MeanAccuracy, color = Task)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) + # Separate lines for each Task
  geom_text(aes(label = Word), vjust = -0.5, hjust = 1.5) + # Task color inherited from ggplot()
  theme(
    legend.position = "top",
    legend.direction = "horizontal"
    # legend.title = element_text(size = 10),
    # legend.text = element_text(size = 5)
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
# ggsave("../graphs/total_accXrt.pdf",width = 8, height = 4)
```

```
mdcen <- d %>%
  filter(Word %in% concrete_accuracy$Word) %>%
  # filter(Experiment == "Ex2: Conc-Abs") %>%
  mutate(Task = ifelse(Task == "Valence", "Valence", "Conceptual"),
    Task = factor(Task, levels = c("Valence", "Conceptual"))) %>%
  mutate(Word = as.factor(Word),
    ID.true = as.factor(ID.true),
    Task = as.factor(Task),
    cAccuracy = as.numeric(Accuracy) - mean(as.numeric(Accuracy)),
    cTask = as.numeric(Task) - mean(as.numeric(Task)))

m <- lmer(LogReactionTime ~ cAccuracy*cTask + (1+cTask|Word) + (1+cTask|ID.true), data = dcen)
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cAccuracy * cTask + (1 + cTask | Word) + (1 +
## cTask | ID.true)
## Data: dcen
##
## REML criterion at convergence: 7409.3
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -9.8359 -0.5999 -0.1593 0.4131 9.5533
##
```

```
## Random effects:
##   Groups   Name                Variance Std.Dev. Corr
##   Word     (Intercept) 0.003690 0.06075
##           cTask        0.006214 0.07883 0.00
##   ID.true  (Intercept) 0.047166 0.21718
##           cTask        0.028544 0.16895 -0.08
##   Residual                0.120935 0.34776
## Number of obs: 9600, groups: Word, 40; ID.true, 40
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      6.78049    0.03583  44.93308 189.216 < 2e-16 ***
## cAccuracy        -0.14525    0.01678 9520.87440  -8.654 < 2e-16 ***
## cTask            -0.13001    0.03033  52.61728  -4.287 7.76e-05 ***
## cAccuracy:cTask   0.12953    0.03355 9520.62607   3.860 0.000114 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cAccrc cTask
## cAccuracy    -0.003
## cTask        -0.063 -0.022
## cAccrcy:cTs  -0.009  0.310 -0.007
```

First Remove participants who aren't super , aggregating over Task

```
length(unique(d$ID.true))

## [1] 40

inacc.parts <- d %>%
  group_by(ID.true) %>%
  summarise(MeanAccuracy = mean(Accuracy)) %>%
  filter(MeanAccuracy < .75)

# How many participants have Accuracy < .75?
length(unique(inacc.parts$ID.true))

## [1] 0

d.inaccurate.removed <- d %>%
  anti_join(inacc.parts, by = "ID.true")

# Sanity check
length(unique(d.inaccurate.removed$ID.true))

## [1] 40

# remove all inaccurate trials
orig <- nrow(d.inaccurate.removed)
d.inaccurate.removed <- d.inaccurate.removed %>%
  filter(Accuracy == 1)
nrow(d.inaccurate.removed)/orig*100

## [1] 94.19792
```

```
# Remove subjects with ReactionTime higher than 3x IQR
summary(d.inaccurate.removed$LogReactionTime)
```

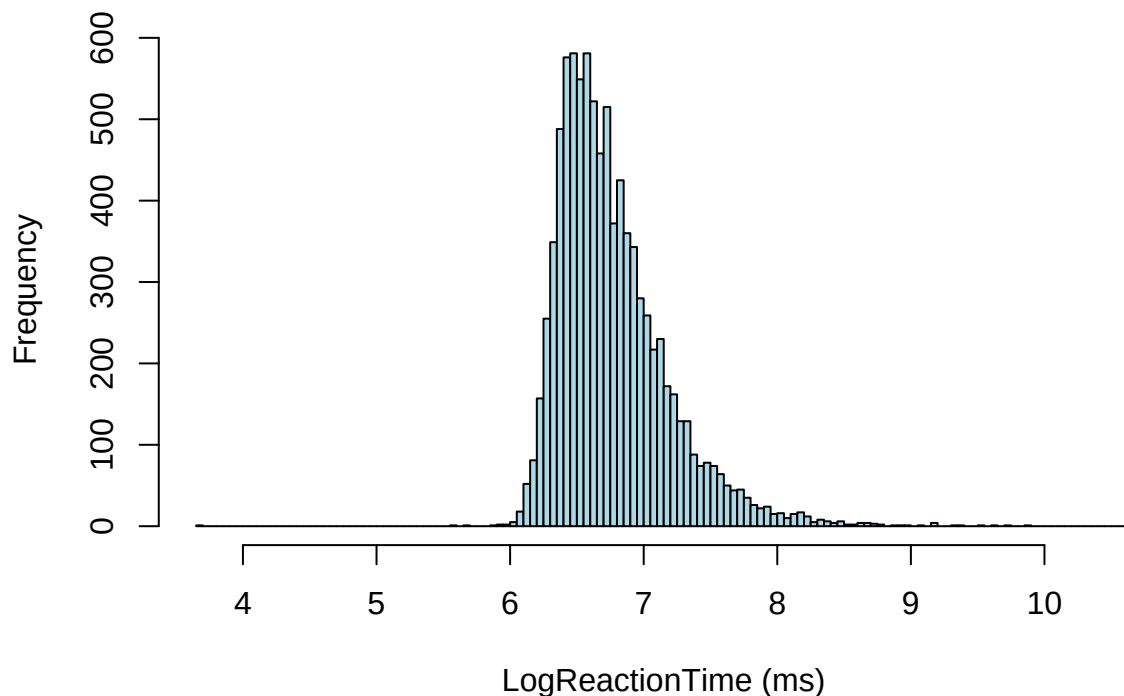
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  3.664   6.472   6.683   6.767   6.964  10.619
```

```
#  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
# 6.924   7.328   7.436   7.479   7.579  10.008
range(d.inaccurate.removed$LogReactionTime)
```

```
## [1]  3.663562 10.618714
```

```
hist(d.inaccurate.removed$LogReactionTime, breaks=100, col="lightblue", xlab="LogReactionTime (ms)",
     main="Histogram with Normal Curve")
```

Histogram with Normal Curve



```
quantile(d.inaccurate.removed$LogReactionTime)
```

```
##      0%      25%      50%      75%     100%
## 3.663562 6.472346 6.683361 6.964136 10.618714
```

```
IQR(d.inaccurate.removed$LogReactionTime)*3 # 0.7526289
```

```
## [1] 1.475368
```

```
cutoff.high <- quantile(d.inaccurate.removed$LogReactionTime)[4] + IQR(d.inaccurate.removed$LogReactionTime)
cutoff.low <- quantile(d.inaccurate.removed$LogReactionTime)[2] - IQR(d.inaccurate.removed$LogReactionTime)
```

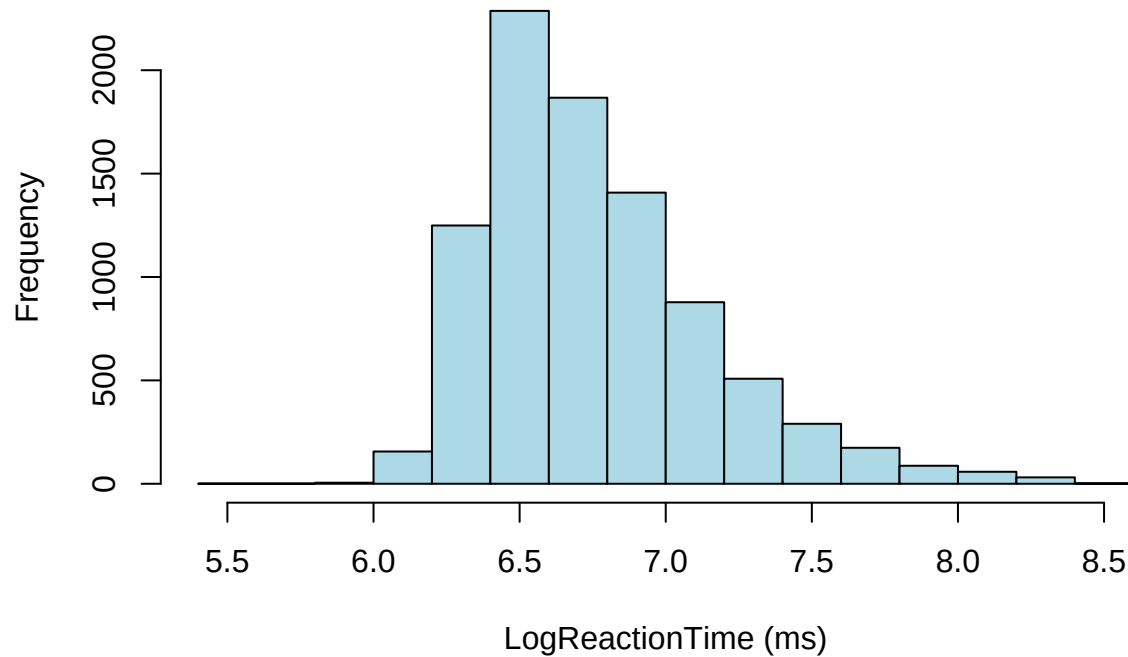
```
# remove subjects with ReactionTime higher than 3 x IQR
```

```
df.outliers.removed <- subset(d.inaccurate.removed, (d.inaccurate.removed$LogReactionTime > cutoff.low)
```

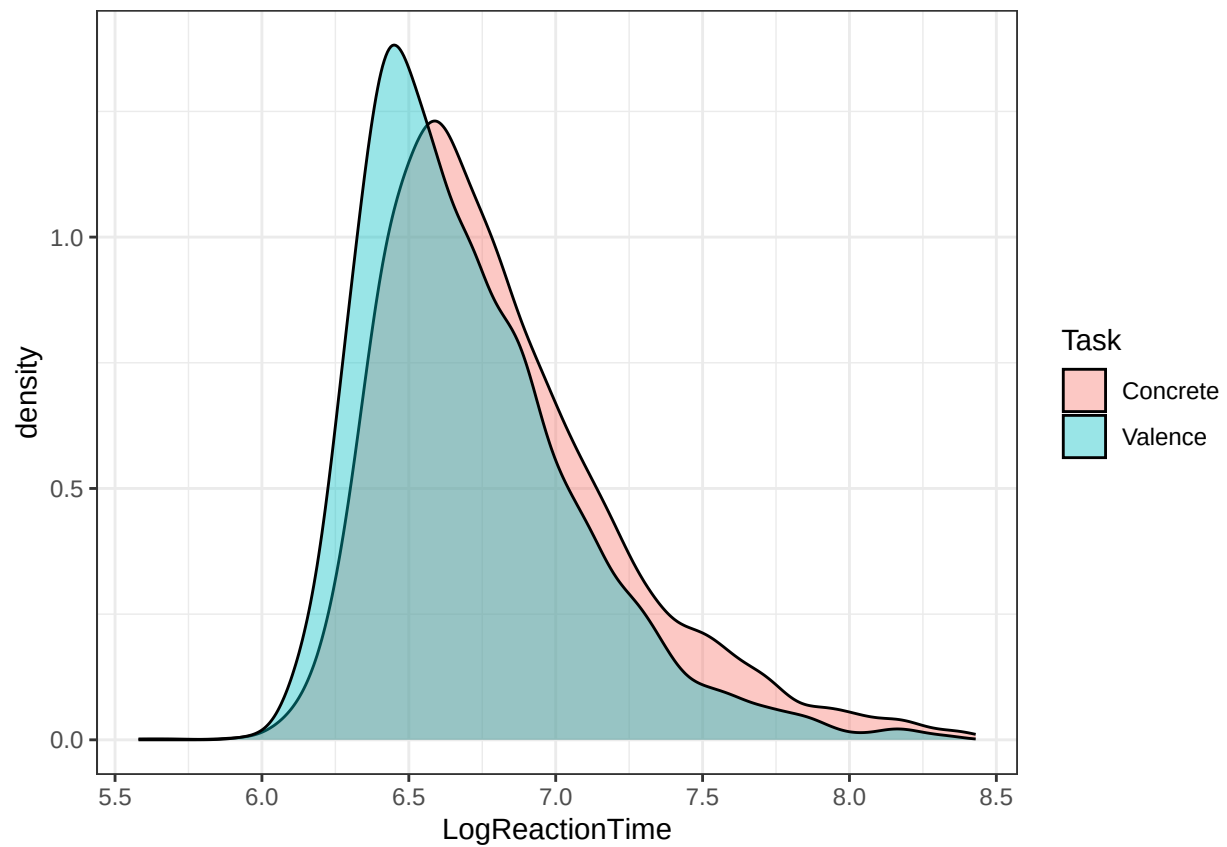
```
hist(df.outliers.removed$LogReactionTime, col="lightblue", xlab="LogReactionTime (ms)",
```

```
main="Histogram with Normal Curve")
```

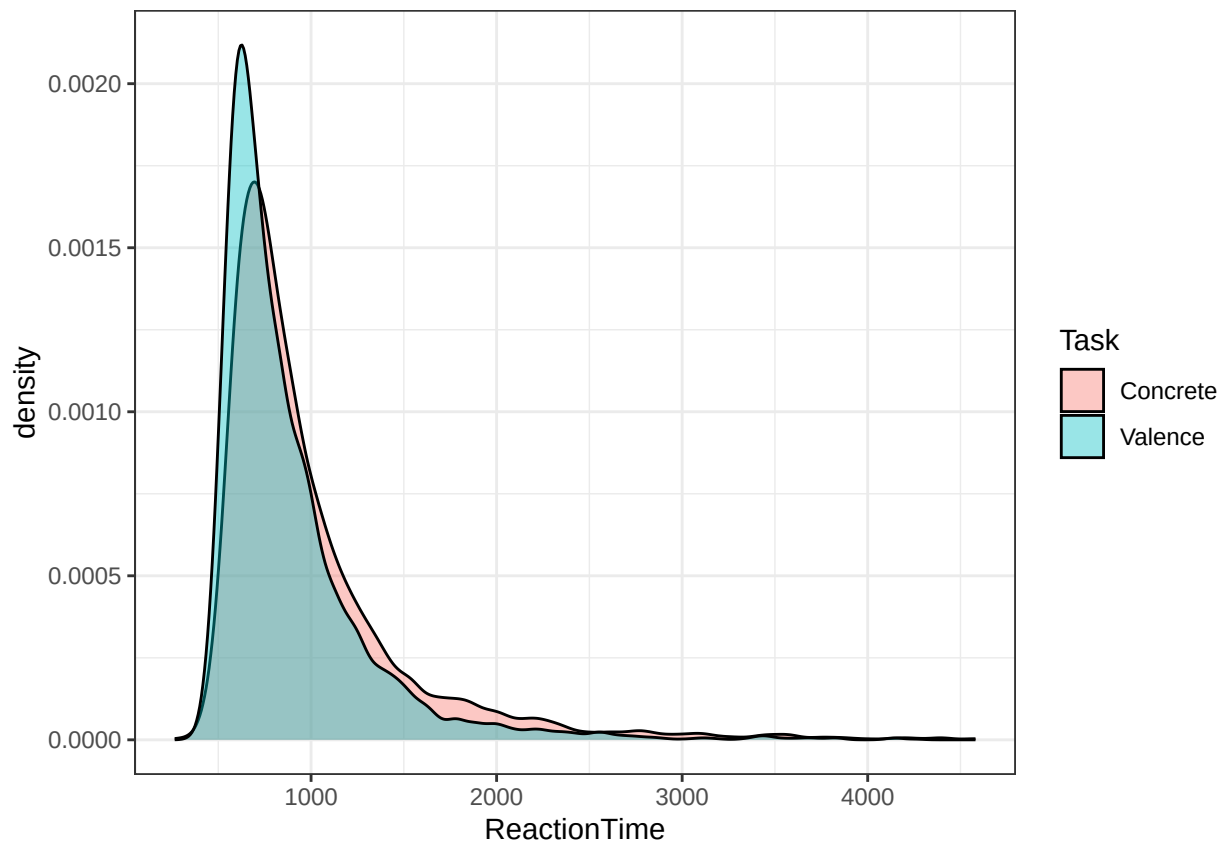
Histogram with Normal Curve



```
ggplot(df.outliers.removed, aes(x=LogReactionTime, fill=Task)) +  
  # facet_wrap(~BlockOrder) +  
  geom_density(alpha = .4)
```



```
ggplot(df.outliers.removed, aes(x=ReactionTime, fill=Task)) +  
  # facet_wrap(~BlockOrder) +  
  geom_density(alpha = .4)
```



convert everything to factors

Is there a difference between Semantic and Valence Tasks?

Yes

```
m = lmer(LogReactionTime ~ cTask + (1+cTask|ID.true) + (1+cTask|Word), data=center)
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cTask + (1 + cTask | ID.true) + (1 + cTask |
##      Word)
##      Data: center
##
## REML criterion at convergence: 4949.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.4410 -0.6368 -0.1738  0.4459  5.1267
##
## Random effects:
##      Groups      Name              Variance Std.Dev. Corr
## ID.true  (Intercept)  0.042593  0.20638
##          cTask       0.021685  0.14726  0.00
## Word    (Intercept)  0.003296  0.05741
##          cTask       0.006609  0.08129  0.04
```

```
## Residual                0.096594 0.31080
## Number of obs: 9002, groups: ID.true, 40; Word, 40
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  6.76237    0.03403 44.87210   198.7 < 2e-16 ***
## cTask        -0.11779    0.02740 56.50604    -4.3 6.84e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## cTask 0.008
```

Is there an Interaction between Task and WordType (ConcVal-Combo)?

Y.

```
m = lmer(LogReactionTime ~ cTask*ConcValCombo + (1+ConcValCombo+cTask|ID.true) + (1+cTask|Word), data=c
# saveRDS(m, "../models/model-Task-ConcValCombo_outlier_excl_ReactionTime_corrected.rds")

# m <- readRDS("../models/model-Task-ConcValCombo_outlier_excl_ReactionTime_corrected.rds")
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cTask * ConcValCombo + (1 + ConcValCombo +
##      cTask | ID.true) + (1 + cTask | Word)
##      Data: center
##
## REML criterion at convergence: 4855.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.4220 -0.6385 -0.1727  0.4445  5.3998
##
## Random effects:
##      Groups      Name                Variance Std.Dev. Corr
## ID.true   (Intercept)                0.046069 0.21464
##           ConcValComboabstract-positive 0.002303 0.04799  -0.04
##           ConcValComboconcrete-negative 0.007090 0.08420  -0.34  0.43
##           ConcValComboconcrete-positive 0.007089 0.08420  -0.25  0.27  0.94
##           cTask                      0.022502 0.15001  -0.12  0.58  0.53
## Word      (Intercept)                0.001980 0.04450
##           cTask                      0.006300 0.07937   0.45
## Residual                        0.094740 0.30780
##
##
##
##
##      0.24
##
```

```
##
##
## Number of obs: 9002, groups: ID.true, 40; Word, 40
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)      6.805942   0.037326  50.763795 182.337
## cTask            -0.156566   0.036971  67.999578  -4.235
## ConcValComboabstract-positive -0.008891   0.023203  41.290061  -0.383
## ConcValComboconcrete-negative -0.091325   0.025662  52.538448  -3.559
## ConcValComboconcrete-positive -0.069527   0.025688  52.532509  -2.707
## cTask:ConcValComboabstract-positive  0.006013   0.039989  35.589586   0.150
## cTask:ConcValComboconcrete-negative  0.068828   0.040013  35.672774   1.720
## cTask:ConcValComboconcrete-positive  0.072813   0.040085  35.926127   1.816
##
##              Pr(>|t|)
## (Intercept)      < 2e-16 ***
## cTask            7.03e-05 ***
## ConcValComboabstract-positive    0.70356
## ConcValComboconcrete-negative    0.00080 ***
## ConcValComboconcrete-positive    0.00915 **
## cTask:ConcValComboabstract-positive  0.88133
## cTask:ConcValComboconcrete-negative  0.09407 .
## cTask:ConcValComboconcrete-positive  0.07765 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cTask  CncVlCmbb- CncVlCmbcnrt-n CncVlCmbcnrt-p
## cTask              0.044
## CncVlCmbb-        -0.290 -0.058
## CncVlCmbcnrt-n    -0.411  0.014  0.478
## CncVlCmbcnrt-p    -0.368 -0.085  0.451    0.619
## cTsk:CncVlCmbb-    -0.104 -0.544  0.338    0.151    0.151
## cTsk:CncVlCmbcnrt-n -0.103 -0.544  0.167    0.306    0.150
## cTsk:CncVlCmbcnrt-p -0.103 -0.543  0.166    0.150    0.303
##
##              cTsk:CncVlCmbb- cTsk:CncVlCmbcnrt-n
## cTask
## CncVlCmbb-
## CncVlCmbcnrt-n
## CncVlCmbcnrt-p
## cTsk:CncVlCmbb-
## cTsk:CncVlCmbcnrt-n  0.502
## cTsk:CncVlCmbcnrt-p  0.501    0.501
```

Main Effect of Block Order

On ReactionTime

- No.

```
m = lmer(LogReactionTime ~ cBlockOrder + (1|ID.true) + (1+cBlockOrder|Word), data=center)
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cBlockOrder + (1 | ID.true) + (1 + cBlockOrder |
##      Word)
##      Data: center
##
## REML criterion at convergence: 5664.1
##
## Scaled residuals:
##      Min      1Q  Median      3Q      Max
## -2.9475 -0.6627 -0.1898  0.4584  5.2991
##
## Random effects:
##      Groups   Name                Variance Std.Dev. Corr
##      ID.true  (Intercept) 0.0411503 0.20286
##      Word     (Intercept) 0.0032216 0.05676
##              cBlockOrder 0.0002784 0.01668 -0.20
##      Residual                0.1065894 0.32648
## Number of obs: 9002, groups:  ID.true, 40; Word, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  6.76006    0.03348 43.77160 201.889  <2e-16 ***
## cBlockOrder -0.08936    0.06457 38.11958  -1.384    0.174
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cBlockOrder -0.002
```

effect of ConcValCombo on ReactionTime?

nope.

```
m = lmer(LogReactionTime ~ ConcValCombo + (1+ConcValCombo|ID.true) + (1|Word), data=center)
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ ConcValCombo + (1 + ConcValCombo | ID.true) +
##      (1 | Word)
##      Data: center
##
## REML criterion at convergence: 5604.6
##
## Scaled residuals:
##      Min      1Q  Median      3Q      Max
## -3.0391 -0.6618 -0.1824  0.4553  5.5499
##
## Random effects:
##      Groups   Name                Variance Std.Dev. Corr
##      ID.true  (Intercept) 0.0441056 0.21001
```

```
##          ConcValComboabstract-positive 0.0007609 0.02758 0.18
##          ConcValComboconcrete-negative 0.0051704 0.07191 -0.28 0.18
##          ConcValComboconcrete-positive 0.0055897 0.07476 -0.21 0.00 0.95
## Word      (Intercept)                  0.0019771 0.04447
## Residual                                0.1052272 0.32439
## Number of obs: 9002, groups: ID.true, 40; Word, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      6.799018   0.036717 51.075736 185.172 < 2e-16
## ConcValComboabstract-positive -0.002779   0.022541 36.746648 -0.123 0.90255
## ConcValComboconcrete-negative -0.084016   0.024875 47.940709 -3.378 0.00146
## ConcValComboconcrete-positive -0.066836   0.025115 48.904312 -2.661 0.01050
##
## (Intercept)          ***
## ConcValComboabstract-positive
## ConcValComboconcrete-negative **
## ConcValComboconcrete-positive *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) CncVlCmbb- CncVlCmbcncrt-n
## CncVlCmbbs-      -0.266
## CncVlCmbcncrt-n -0.385 0.454
## CncVlCmbcncrt-p -0.355 0.433      0.597
```

In the Concreteness task, is there a difference between concreteness and abstractness on ReactionTime?

- Nope

```
str(df_factors)
```

```
## 'data.frame': 9002 obs. of 20 variables:
## $ X : Factor w/ 9002 levels "1","2","3","4",...: 1 2 3 4 5 6 7 8 9 10 ...
## $ ID.true : Factor w/ 40 levels "558a035bdf99b2d75651378",...: 14 14 14 14 14 14 14 14 14 14 ...
## $ Word : Factor w/ 40 levels "admired","annoy",...: 25 4 32 14 39 33 7 28 29 5 ...
## $ Label : Factor w/ 2 levels "test_conc","test_val": 1 1 1 1 1 1 1 1 1 1 ...
## $ ConcValCombo : Factor w/ 4 levels "abstract-negative",...: 3 2 3 1 3 3 2 4 4 3 ...
## $ Task : Factor w/ 2 levels "Concrete","Valence": 1 1 1 1 1 1 1 1 1 1 ...
## $ BlockOrder : Factor w/ 2 levels "CV","VC": 1 1 1 1 1 1 1 1 1 1 ...
## $ Group : Factor w/ 2 levels "A","B": 2 2 2 2 2 2 2 2 2 2 ...
## $ Response : Factor w/ 4 levels "abstract","concrete",...: 2 1 2 1 2 2 1 2 2 2 ...
## $ Accuracy : Factor w/ 1 level "1": 1 1 1 1 1 1 1 1 1 1 ...
## $ EventTime : Factor w/ 9002 levels "1732203168805",...: 6512 6513 6514 6515 6516 6517 6518 6519 ...
## $ Value : Factor w/ 4 levels "abstract","concrete",...: 2 1 2 1 2 2 1 2 2 2 ...
## $ RT : Factor w/ 1385 levels "1270","1287.833333333333",...: 311 990 749 385 531 149 50 5 ...
## $ ReactionTime : int 965 1809 1020 799 715 754 668 1252 719 1205 ...
## $ Key_value_F : Factor w/ 4 levels "abstract","concrete",...: 2 2 2 2 2 2 2 2 2 2 ...
## $ Key_value_J : Factor w/ 2 levels "A","B": 2 2 2 2 2 2 2 2 2 2 ...
## $ Comments : Factor w/ 0 levels: NA NA NA NA NA NA NA NA NA NA ...
## $ LogReactionTime: num 6.87 7.5 6.93 6.68 6.57 ...
## $ LogRT : Factor w/ 1385 levels "7.14677217945264",...: 311 990 749 385 531 149 50 542 662 ...
```

```
## $ TrialNumber : Factor w/ 240 levels "1","2","3","4",...: 1 2 3 4 8 9 10 11 12 14 ...
```

```
sem <- df_factors %>%
  filter(Task == "Concrete") %>%
  mutate(
    Semantic = ifelse(grepl("concrete", ConcValCombo), "concrete",
                      ifelse(grepl("abstract", ConcValCombo), "abstract", NA)),
    Valence = ifelse(grepl("positive", ConcValCombo), "positive",
                     ifelse(grepl("negative", ConcValCombo), "negative", NA)),
    cConcValCombo = as.numeric(ConcValCombo) - mean(as.numeric(ConcValCombo)),
    cSemantic = as.numeric(factor(Semantic)) - mean(as.numeric(factor(Semantic)))
  )
```

```
m = lmer(LogReactionTime ~ cConcValCombo + (1+cConcValCombo|ID.true) + (1+cConcValCombo|Word), data=sem,
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cConcValCombo + (1 + cConcValCombo | ID.true) +
## (1 + cConcValCombo | Word)
## Data: sem
##
## REML criterion at convergence: 3016.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2380 -0.6385 -0.1703  0.4680  5.1398
##
## Random effects:
## Groups Name Variance Std.Dev. Corr
## ID.true (Intercept) 0.0487592 0.22081
## cConcValCombo 0.0031780 0.05637 -0.16
## Word (Intercept) 0.0022190 0.04711
## cConcValCombo 0.0003197 0.01788 0.58
## Residual 0.1092111 0.33047
## Number of obs: 4396, groups: ID.true, 40; Word, 40
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 6.82551 0.03618 42.86888 188.651 < 2e-16 ***
## cConcValCombo -0.04647 0.01241 45.00765 -3.745 0.00051 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr)
## cConcValCmb -0.062
```

```
m = lmer(LogReactionTime ~ cSemantic + (1+cSemantic|ID.true) + (1|Word), data=sem)
summary(m)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cSemantic + (1 + cSemantic | ID.true) + (1 |
## Word)
```

```
## Data: sem
##
## REML criterion at convergence: 2983
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2902 -0.6387 -0.1801  0.4684  5.1300
##
## Random effects:
##   Groups   Name                Variance Std.Dev. Corr
##   ID.true  (Intercept)  0.048623  0.22051
##           cSemantic    0.018249  0.13509  -0.21
##   Word     (Intercept)  0.001838  0.04287
##   Residual                0.108476  0.32936
## Number of obs: 4396, groups: ID.true, 40; Word, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  6.82580    0.03587 41.76458 190.290 < 2e-16 ***
## cSemantic   -0.11175    0.02721 53.30476  -4.106 0.000139 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cSemantic -0.163
```

In the Valence task , is there a difference between positive and negative on ReactionTime?

- Nope.

```
val <- df_factors %>%
  filter(Task == "Valence") %>%
  mutate(
    Semantic = ifelse(grepl("concrete", ConcValCombo), "concrete",
                      ifelse(grepl("abstract", ConcValCombo), "abstract", NA)),
    Valence = ifelse(grepl("positive", ConcValCombo), "positive",
                     ifelse(grepl("negative", ConcValCombo), "negative", NA)),
    cConcValCombo = as.numeric(ConcValCombo) - mean(as.numeric(ConcValCombo)),
    cValence = as.numeric(factor(Valence)) - mean(as.numeric(factor(Valence)))
  )

m = lmer(LogReactionTime ~ cConcValCombo + (1+cConcValCombo|ID.true) + (1+cConcValCombo|Word), data=val)

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 0.00331201 (tol = 0.002, component 1)

summary(m)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cConcValCombo + (1 + cConcValCombo | ID.true) +
##          (1 + cConcValCombo | Word)
```

```

## Data: val
##
## REML criterion at convergence: 1756.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.9001 -0.6409 -0.1724  0.4253  5.5412
##
## Random effects:
##   Groups   Name                Variance Std.Dev. Corr
##   ID.true  (Intercept)         0.0479600 0.21900
##           cConcValCombo 0.0003191 0.01786  0.08
##   Word     (Intercept)         0.0040271 0.06346
##           cConcValCombo 0.0008182 0.02860  0.27
##   Residual                        0.0807067 0.28409
## Number of obs: 4606, groups: ID.true, 40; Word, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   6.70438    0.03660 46.62800 183.170  <2e-16 ***
## cConcValCombo -0.01744    0.01153 26.90471  -1.513   0.142
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cConcValCmb 0.069
## optimizer (nlptwrap) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 0.00331201 (tol = 0.002, component 1)
m = lmer(LogReactionTime ~ cValence + (1+cValence|ID.true) + (1|Word), data=val)
summary(m)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: LogReactionTime ~ cValence + (1 + cValence | ID.true) + (1 |
##      Word)
## Data: val
##
## REML criterion at convergence: 1759.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.9972 -0.6376 -0.1684  0.4222  5.6141
##
## Random effects:
##   Groups   Name                Variance Std.Dev. Corr
##   ID.true  (Intercept)         0.048008 0.21911
##           cValence           0.001278 0.03575  0.15
##   Word     (Intercept)         0.005371 0.07328
##   Residual                        0.080784 0.28423
## Number of obs: 4606, groups: ID.true, 40; Word, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)

```

```

## (Intercept)  6.705000    0.036770 47.318316 182.348    <2e-16 ***
## cValence      0.008149    0.025286 40.503327   0.322     0.749
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr)
## cValence 0.032

```